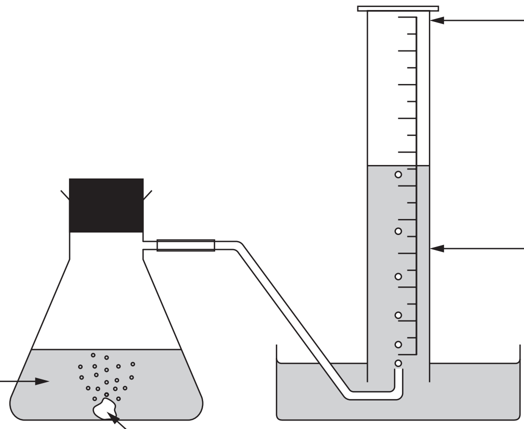
Examiner only

2. A student carried out an experiment to investigate the speed of the reaction between a **lump** of magnesium carbonate of mass 0.32 g and excess dilute hydrochloric acid at 20 °C.

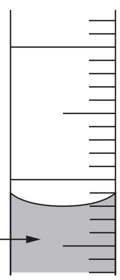
Magnesium carbonate reacts with dilute hydrochloric acid forming carbon dioxide gas. The total volume of carbon dioxide formed was recorded every 5 minutes for 40 minutes.



The results are shown below. The result for 15 minutes is missing.

Time (minutes)	0	5	10	15	20	25	30	35	40
Volume of carbon dioxide formed (cm^3)	0	20	41		79	83	90	90	90

(a) Use the diagram below to find the volume of carbon dioxide gas formed after 15 minutes. [1]



volume of carbon dioxide = ... cm^3

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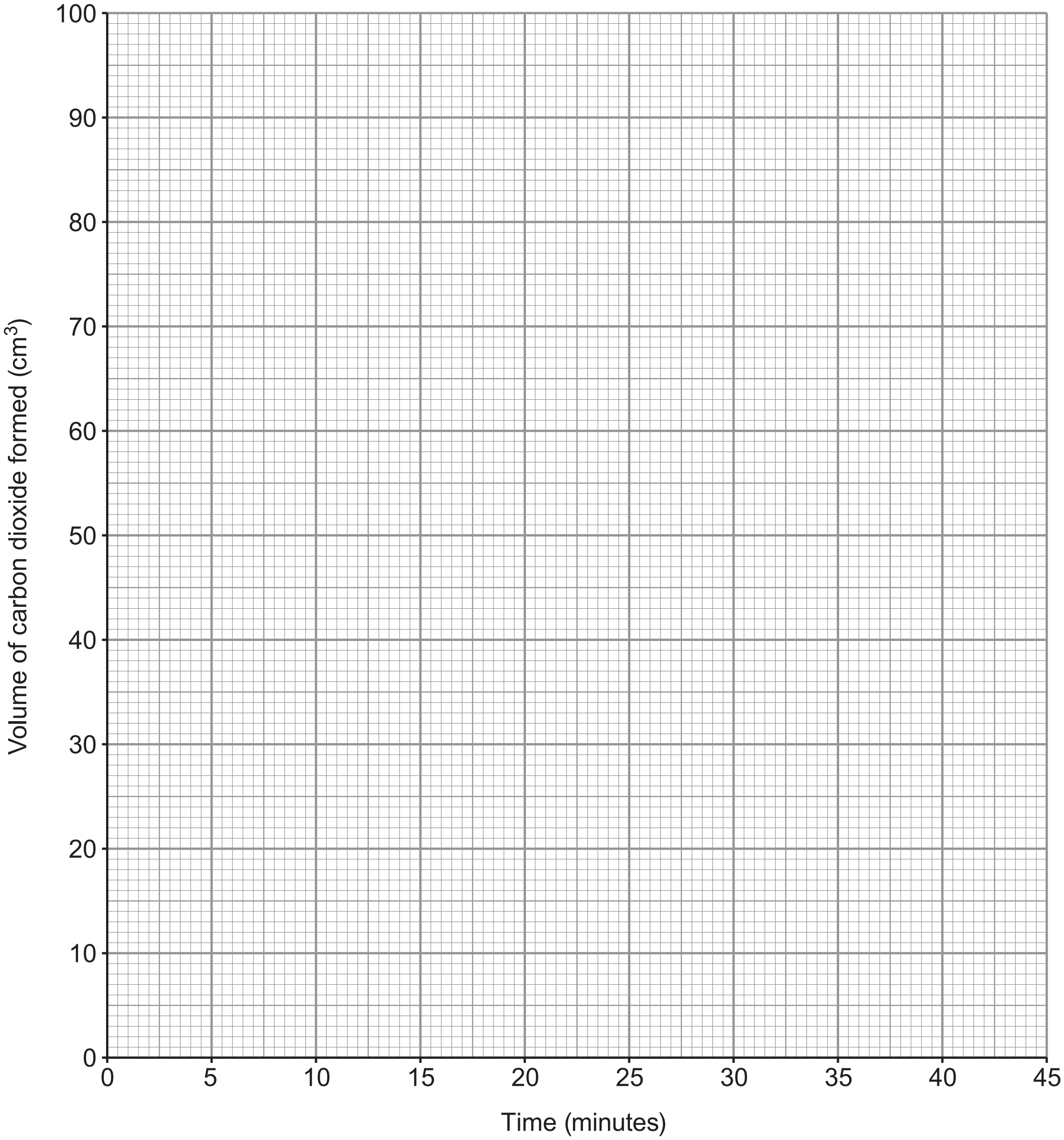
(b) Plot the results from the table, including your answer to part *(a)*, on the grid below. Draw a suitable line and **label this graph X**. [3]



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(c) Sketch the graph you would expect if the experiment were repeated using 0.32 g of magnesium carbonate **powder** instead of the lump of magnesium carbonate.
Label this graph Y. [2]

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(d) State and explain, using particle theory, the effect of **increasing** the concentration of the hydrochloric acid. [3]

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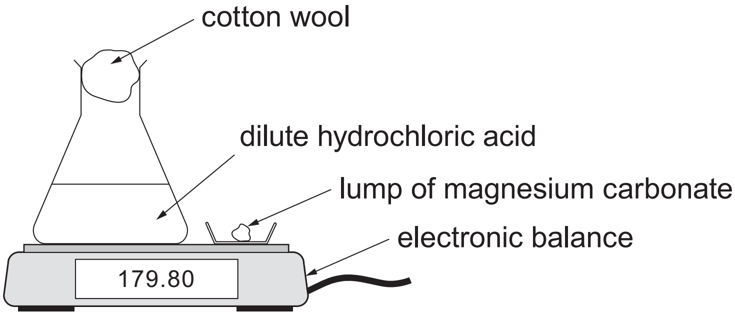
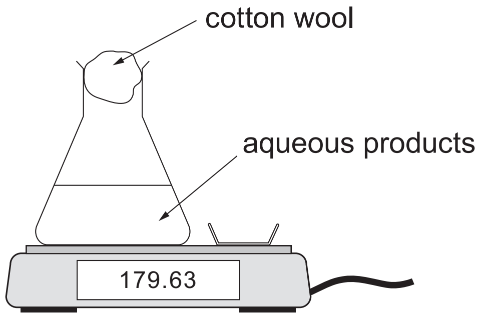
(e) The student investigated the same reaction using a different apparatus. A lump of magnesium carbonate was added to excess dilute hydrochloric acid at 20 °C.



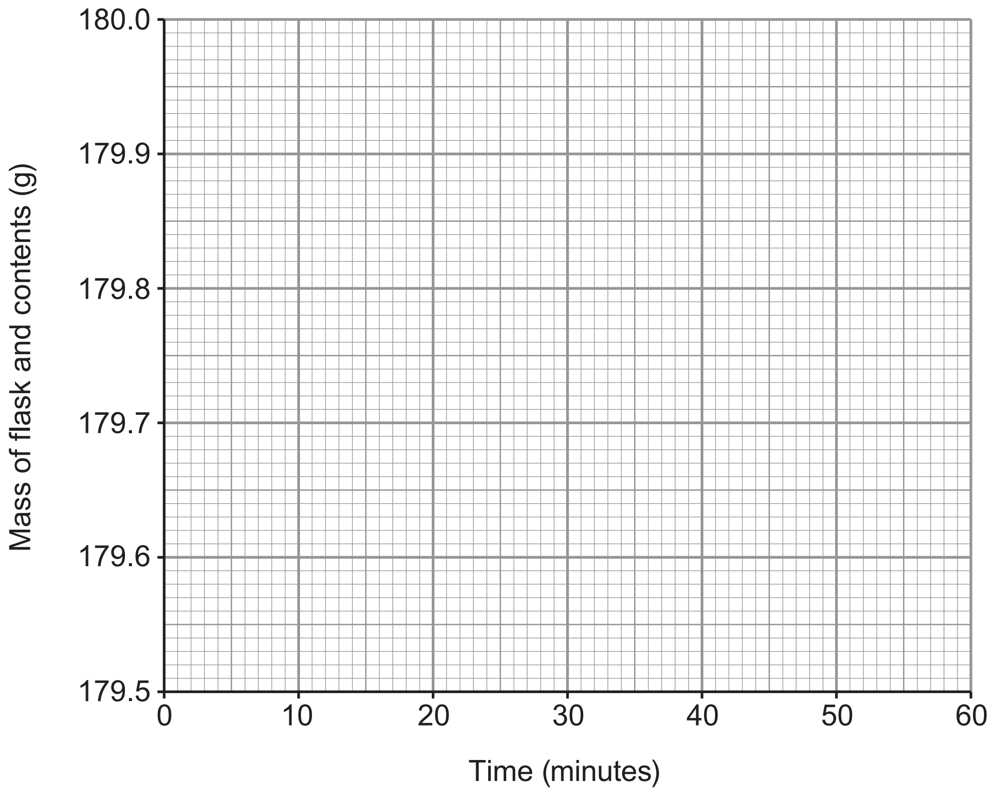
Apparatus at the start



Apparatus after 60 minutes

The change in mass was recorded for 60 minutes and displayed as a graph on a computer screen. The reaction took 40 minutes to complete.

Sketch the graph you would expect to see. [2]



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